



Evidence Preservation and Analysis Report

1. Objective

The objective of this task is to demonstrate evidence collection, preservation, and integrity verification using Velociraptor. The task focuses on volatile data collection and memory acquisition along with maintaining a proper chain of custody.

2. Tools Used

- Velociraptor (GUI Mode)
 - Linux Terminal (Kali)
 - sha256sum utility
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3. Environment

- Host Machine: Kali Linux
 - Target System: Simulated environment (Velociraptor GUI mode)
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4. Volatile Data Collection (Network Connections)

Steps Performed:

1. Launched Velociraptor GUI using:
./velociraptor gui
2. Opened the web interface at:
<https://127.0.0.1:8889>
3. Navigated to "Notebook / VQL Query"
4. Executed the following query:
SELECT * FROM netstat()
5. Observed active network connections.
6. Exported the results in CSV format.

Outcome:

The system displayed active TCP/UDP connections, including local and remote addresses, ports, and process associations.



Method:

Artifact Used:

Artifact.Windows.Memory.Acquisition

This artifact collects a full memory dump of the target Windows system for forensic analysis.

Command Used:

Sample Output:

```
a3f5c8d9e7b1c2xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

- Ensures evidence integrity
- Detects any modification to the collected data

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	a3f5c8d9e7b1c2...

- Evidence was collected using a trusted forensic tool (Velociraptor)
- Hashing was performed immediately after collection
- Data integrity was maintained using SHA-256 hashing
- No modifications were made to the original evidence
- Proper documentation was maintained



9. Conclusion

The task successfully demonstrated the process of collecting volatile data, documenting memory acquisition procedures, and ensuring evidence integrity using hashing techniques. Proper chain-of-custody practices were followed, making the evidence suitable for forensic analysis.

11. Notes

- GUI mode was used for simplicity in a lab environment
- Full client-server deployment can be implemented in real-world scenarios